

Supplementary Material: Operator-Specific Forgetting in Matrix-Valued Associative Memory

Javier Escobar
Independent Researcher

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1 A. Reproducibility manifest

Reference commit: `cec4ffe`. Tag: `lab-radioactivo-v1`.

The repository contains implementations, execution runners, evaluation probes, audit logs, model checkpoints, frozen claims, and execution transcripts for experimental milestones O01–O05. The primary trained-model gate is O05. Verification tooling:

- `paper/verify_numbers.py`: Recomputes all quantitative values and verification gates directly from frozen JSON artifacts.
- `paper/make_figures.py`: Regenerates all publication figures (PDF and PNG at 300 DPI) from the same frozen artifacts.

2 B. Notation and memory construction

D denotes the key and value dimension ($D \in \{64, 128\}$). n is the number of stored key–value pairs. $c = (n - 1)/D$ is the associative crosstalk load excluding the target item. $\mathbf{w}_i \in \mathbb{C}^D$ are fixed codebook keys with unit-modulus entries ($|\mathbf{w}_{i,k}| = 1$, squared Euclidean norm $\|\mathbf{w}_i\|_2^2 = D$). $\mathbf{v}_i \in \mathbb{R}^D$ are real-valued target vectors. The memory state $\mathbf{M} \in \mathbb{C}^{D \times D}$ stores associations via matrix superposition:

$$\mathbf{M} = \sum_{i=1}^n \mathbf{w}_i \mathbf{v}_i^\top. \quad (1)$$

The matched-filter query read using key $\mathbf{w}_j$ is computed as $\hat{\mathbf{v}}_j = \mathbf{w}_j^H \mathbf{M} / D$. The pooled residual metric is defined as

$$L = \frac{\sum_i \|\mathbf{r}_i\|^2}{\sum_i \|\mathbf{v}_i\|^2}, \quad (2)$$

where $\mathbf{r}_i$ is the post-erase query readout residual.

3 C. Formal operator definitions

3.1 C.1 Key drift reread erase

Memory items are stored unperturbed. At erase time, local oscillator drift alters the erase key to $\tilde{\mathbf{w}}_j = \mathbf{w}_j e^{i\delta}$. The erase operator re-reads using $\tilde{\mathbf{w}}_j$ and subtracts the outer product:

$$\hat{\mathbf{v}}_j = \frac{\tilde{\mathbf{w}}_j^H \mathbf{M}}{D}, \quad \mathbf{M}' = \mathbf{M} - \tilde{\mathbf{w}}_j \hat{\mathbf{v}}_j. \quad (3)$$

Because both the projection estimate and the subtractive unbinding use $\tilde{\mathbf{w}}_j$, the operation represents an exact orthogonal projection onto the subspace orthogonal to $\tilde{\mathbf{w}}_j$.

3.2 C.2 State drift reread erase

We use state drift to denote a write-time phase offset attached to the target association, followed by a mismatch between the stored phase coordinate used for re-reading and the reference coordinate used for unbinding. Specifically, the target association is stored as $(e^{i\delta}\mathbf{w}_j)\mathbf{v}_j^\top$. At erase time, the erase operator re-reads using the item's stored key coordinate $\tilde{\mathbf{w}}_j = e^{i\delta}\mathbf{w}_j$ to recover $\hat{\mathbf{v}}_j = \tilde{\mathbf{w}}_j^H \mathbf{M}/D$, but subtracts using the current reference clock key $\mathbf{w}_j$:

$$\mathbf{M}' = \mathbf{M} - \mathbf{w}_j \hat{\mathbf{v}}_j. \quad (4)$$

3.3 C.3 Real-truncated readout channel

When evaluating the real channel, real truncation is applied to the re-read estimate before subtraction: $\hat{\mathbf{v}}_j \leftarrow \text{Re}(\hat{\mathbf{v}}_j)$. At query time, the readout residual is evaluated as $\mathbf{r}_j = \text{Re}(\mathbf{w}_j^H \mathbf{M}'/D)$.

3.4 C.4 Representational erase control

The erased value is re-presented from an external target rather than re-read from the state: $\mathbf{M}' = \mathbf{M} - \mathbf{w}_j \mathbf{v}_j^\top$. This serves as a control condition separating operator-level self-estimation from state residual dynamics.

3.5 C.5 Normalized delta control

The corrective delta-rule update modifies state $\mathbf{S} \in \mathbb{R}^{D \times D}$ via

$$\mathbf{S}_t = \mathbf{S}_{t-1} + \beta \mathbf{k}_t (\mathbf{v}_t - \mathbf{k}_t^\top \mathbf{S}_{t-1})^\top. \quad (5)$$

In the normalized-key delta control arm, keys are normalized ($\|\mathbf{k}_t\|_2^2 = 1$). For $0 < \beta < 2$, this maintains the directional feedback multiplier $|1 - \beta| < 1$ strictly inside the stable unit disk.

4 D. Closed-form law derivations

4.1 D.1 Distributed crosstalk scaling

For independent random unit-modulus keys $\mathbf{w}_i, \mathbf{w}_k \in \mathbb{C}^D$, the inner products satisfy $\mathbb{E}[|\mathbf{w}_i^H \mathbf{w}_k|^2] = D$ for $i \neq k$. After normalization by D , each interfering association contributes expected energy $\mathbb{E}[|\mathbf{w}_i^H \mathbf{w}_k/D|^2] = 1/D$. Summing over $n - 1$ interfering associations yields the crosstalk load $c = (n - 1)/D$, consistent with distributed associative memory capacity [1–3].

4.2 D.2 Key drift under complex channel

The operator $\mathbf{P}_{\tilde{\mathbf{w}}_j}^\perp = \mathbf{I} - \tilde{\mathbf{w}}_j \tilde{\mathbf{w}}_j^H / D$ is an exact rank-1 projection matrix. When querying with reference key $\mathbf{w}_j = \tilde{\mathbf{w}}_j e^{-i\delta}$, the target component in $\mathbf{M}'$ along direction $\tilde{\mathbf{w}}_j$ is identically zero. Therefore, $L_{\text{key, complex}}(\delta) = 0$ in the algebraic model.

4.3 D.3 State drift under complex channel

With target item stored as $(e^{i\delta}\mathbf{w}_j)\mathbf{v}_j^\top$ and unbinding performed with reference key $\mathbf{w}_j$, the re-read estimate is:

$$\hat{\mathbf{v}}_j = \frac{\tilde{\mathbf{w}}_j^H \mathbf{M}}{D} = e^{-i\delta} \mathbf{v}_j^\top + \frac{e^{-i\delta}}{D} \sum_{k \neq j} \mathbf{w}_j^H \mathbf{w}_k \mathbf{v}_k^\top. \quad (6)$$

Subtracting $\mathbf{w}_j \hat{\mathbf{v}}_j$ yields the post-erase state:

$$\mathbf{M}' = (e^{i\delta} - 1) \mathbf{w}_j \mathbf{v}_j^\top + \sum_{k \neq j} \mathbf{w}_k \mathbf{v}_k^\top - \mathbf{w}_j \frac{e^{-i\delta}}{D} \sum_{k \neq j} \mathbf{w}_j^H \mathbf{w}_k \mathbf{v}_k^\top. \quad (7)$$

Reading with reference key $\mathbf{w}_j$ yields target residual $(e^{i\delta} - 1)\mathbf{v}_j^\top$ and interference residual

$$\mathbf{r}_{\text{cross}} = (1 - e^{-i\delta}) \frac{1}{D} \sum_{k \neq j} (\mathbf{w}_j^H \mathbf{w}_k) \mathbf{v}_k^\top. \quad (8)$$

Taking expected squared norms over independent unit-modulus keys gives

$$\mathbb{E}[\|\mathbf{r}_j\|^2] = |e^{i\delta} - 1|^2 \|\mathbf{v}_j\|^2 + |1 - e^{-i\delta}|^2 \frac{n-1}{D} \|\mathbf{v}_j\|^2 = 2(1 - \cos \delta)(1 + c) \|\mathbf{v}_j\|^2. \quad (9)$$

Because the factor $2(1 - \cos \delta)(1 + c)$ is invariant across target vectors, the ratio of the sum of expected residual energies to the sum of target energies preserves the identical pooled law:

$$\frac{\sum_j \mathbb{E}[\|\mathbf{r}_j\|^2]}{\sum_j \|\mathbf{v}_j\|^2} = 2(1 - \cos \delta)(1 + c). \quad (10)$$

4.4 D.4 Real-truncated channel derivations

Applying real projection $\text{Re}(\cdot)$ both at the re-read estimate $\hat{\mathbf{v}}_j \leftarrow \text{Re}(\hat{\mathbf{v}}_j)$ and at the final query readout $\mathbf{r}_j = \text{Re}(\mathbf{w}_j^H \mathbf{M}'/D)$ projects out the imaginary component of the phase differences.

- **Key drift (Re channel):** Under key drift, the target contribution to the real query residual is $\sin^2 \delta \cdot \mathbf{v}_j^\top$, yielding target energy $\sin^4 \delta \|\mathbf{v}_j\|^2$. The interfering crosstalk from $n-1$ unperturbed associations integrates over independent random phases to $\frac{c}{4}(1 - \cos 2\delta) \|\mathbf{v}_j\|^2$. Cross-terms between target and interference vanish in expectation due to zero-mean key correlations ($\mathbb{E}[\mathbf{w}_j^H \mathbf{w}_k] = 0$ for $k \neq j$). Summing gives:

$$L_{\text{key, Re}}(\delta) = \sin^4 \delta + \frac{c}{4}(1 - \cos 2\delta). \quad (11)$$

- **State drift (Re channel):** The target association written with phase offset $e^{i\delta}$ yields an unbinding target residual $(1 - \cos \delta)\mathbf{v}_j^\top$, with energy $(1 - \cos \delta)^2 \|\mathbf{v}_j\|^2$. The unbinding subtracts the real part of the re-read crosstalk, leaving residual interference energy $c(1 - \cos \delta) \|\mathbf{v}_j\|^2$. Independent key phases ensure vanishing cross-terms in expectation, yielding:

$$L_{\text{state, Re}}(\delta) = (1 - \cos \delta)^2 + c(1 - \cos \delta). \quad (12)$$

5 E. Write stability analysis

Along direction $\mathbf{k}_t$, projecting the state error yields the first-order scalar recursion

$$s_t = (1 - \beta \|\mathbf{k}_t\|^2) s_{t-1} + \beta q_t, \quad (13)$$

where q_t represents cross-talk from other stored patterns. The homogeneous dynamics are stable if and only if $|1 - \beta \|\mathbf{k}_t\|^2| < 1$, which requires $0 < \beta \|\mathbf{k}_t\|^2 < 2$.

With $\beta = 1$ and unnormalized embedding initialization ($\|\mathbf{k}_t\|^2 \approx 8$), the multiplier $|1 - 8| = 7$ induces exponential growth ($\|\mathbf{S}_t\| \sim 7^t$), causing numerical explosion within early epochs ($\|\mathbf{S}\| \approx 10^{11}-10^{12}$). Key normalization ensures $\|\mathbf{k}_t\|_2^2 = 1$; for $0 < \beta < 2$, this yields $|1 - \beta| < 1$ and keeps the directional dynamics inside the unit disk.

6 F. Experimental protocol (O05 milestone)

- **Dimensions:** Primary scale $D = 128$ ($n = 40$ pairs, predicted load $c = 39/128 \approx 0.3047$, five seeds) compared against $D = 64$ ($n = 20$ pairs, predicted load $c = 19/64 \approx 0.2969$, three seeds), achieving approximately matched load (2.7% difference).
- **Vocabulary:** Discrete alphabet of 105 tokens (48 key tokens, 48 value tokens, special marker tokens). The token vocabulary is used strictly to construct controlled key–value associative sequences; the evaluation is not intended as a natural-language modeling benchmark.
- **Arms:** `wave_complex`, `wave_re`, `normalized_delta`.
- **Events and Units:** 320 forgetting events per arm and seed (exceeding pre-registered minimum of 300). For all inferential statistical tests, the random seed was treated as the independent experimental unit.
- **Drift grid:** $\delta \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$.
- **Ridge probe:** Linear Ridge regression ($\alpha = 0.1$, within-model split of 75% train [$n = 240$] / 25% test [$n = 80$]) mapping readout residuals $\mathbf{r}_j$ to continuous targets $\mathbf{v}_j \in \mathbb{R}^D$. Continuous predictions were decoded to the discrete value vocabulary by nearest cosine distance, with exact match recorded when the recovered token matched the true target token.
- **Selectivity:** $\text{Selectivity} = \text{EM}_{\text{alive}} - \text{EM}_{\text{erased}}$.
- **Equivalence test:** Two one-sided t -test (TOST at $\alpha = 0.05$, two-sided 90% CI) on retrieval selectivity between `wave_complex` and `normalized_delta` with pre-registered margin $\varepsilon = 0.02$.

7 G. Numerical results summary

At $D = 128$, all 15 seed–arm combinations reached $\text{EM} = 1.000$; at $D = 64$, all 9 seed–arm combinations likewise trained to $\text{EM} = 1.000$. The four closed-form laws predicted empirical pooled residuals with maximum absolute deviation of 0.0049 at $D = 128$ and 0.0094 at $D = 64$, remaining below the 0.02 verification threshold. Dimension-matched load comparison passed for all four cells with maximum cross-scale difference 0.0064.

In the key/complex condition, all 30 seed $\times$ drift runs produced algebraically zero and numerically null residuals below the numerical guard threshold $\varepsilon_{\text{null}} = 10^{-4}$. TOST yielded mean difference -0.0031 with 90% CI $[-0.0206, 0.0144]$; because the lower bound crosses $-\varepsilon = -0.02$ by 0.0006 under $n = 5$ seeds, the result is reported as inconclusive with respect to formal equivalence.

8 H. Metric audit

A simple mean of per-example ratios $\frac{1}{N} \sum_i (\|\mathbf{r}_i\|^2 / \|\mathbf{v}_i\|^2)$ introduced severe estimation variance in trained models due to small target vector norms in the denominator. The primary metric was therefore standardized as the pooled ratio of sums $L = \sum_i \|\mathbf{r}_i\|^2 / \sum_i \|\mathbf{v}_i\|^2$, stabilizing the estimator.

When the post-erase residual is algebraically zero, computing erased-item exact match by taking an argmax over random noise is uninformative. In our evaluation protocol, residuals below $\varepsilon_{\text{null}} = 10^{-4}$ are recorded as numerically null residuals.

9 I. Negative results and scope boundaries

The vector-state geometry did not support the intended D -scaling law. The ideal wave erase and ideal delta erase were algebraically identical under normalized keys. The unnormalized $\beta = 1$ corrective delta baseline diverged due to unnormalized key norms. Learned codebooks were not included in the closed-form theory. These results define the scope of the present analysis.

References

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- [2] Pentti Kanerva. Hyperdimensional computing: An introduction to computing in distributed representation with high-dimensional random vectors. *Cognitive Computation*, 1(2):139–159, 2009. doi: 10.1007/s12559-009-9009-8.
- [3] Tony A. Plate. Holographic reduced representations. *IEEE Transactions on Neural Networks*, 6(3):623–641, 1995. doi: 10.1109/72.377968.